Maninder Singh

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Education

Indian Institute of Technology (Indian School of Mines) Dhanbad, India
Bachelor of Technology in Electrical Engineering (CGPA: 6.60 / 10.00) *2023 – 2027*

- **Relevant Coursework:** Power Electronics, Modern Control Systems, Electrical Machines, Microcontrollers & Microprocessors, Digital Logic Design, Data Structures & Algorithms

Experience

Mechismu Racing Electric, IIT Dhanbad 2024 – Present
Team Vice Captain *Formula Bharat Electric Vehicle Team*

- Led coordination across multidisciplinary subsystem teams during electric race vehicle development.
- Contributed to **HV/LV electrical architecture, safety electronics, and system integration**.
- Participated in powertrain debugging, subsystem validation, and vehicle testing activities.
- Contributed to the team's **12th rank in Engineering Design** and **24th overall finish** at Formula Bharat 2025.

Projects

Energy-Efficient Six-Phase Induction Motor Control using IRFOC *MATLAB/Simulink, TI TMS320F28379D DSP*
Mentor: *Prof. Sukanta Das (HOD, Electrical Engineering)*
Members: *Maninder Singh, Aditya Krishna*

- Designed a **six-phase Indirect Rotor Field Oriented Control (IRFOC)** architecture for **EV traction applications**.
- Integrated a **Loss Model Controller (LMC)** for online flux optimization to minimize machine losses while preserving torque response.
- Implemented **six-phase Clarke/Park transformations, rotor flux estimation, and closed-loop current control**.
- Deploying and validating the control strategy on **TI TMS320F28379D DSP** for real-time execution and robustness analysis under parameter variations.

Delta-Sigma ADC Design for ISRO VLSI Challenge (Inter-IIT Tech Meet 14.0) *MATLAB/Simulink, Mixed-Signal Modeling*

- Solved the official **ISRO VLSI problem statement** at **Inter-IIT Tech Meet 14.0**, achieving **6th rank among 22 IITs**.
- Designed and evaluated **CT, DT, MASH, SMASH, and Hybrid $\Delta\Sigma$ architectures** under thermal and flicker-noise conditions.
- Optimized a **3rd-order Continuous-Time $\Delta\Sigma$ modulator** (**OSR=128**, **16-level quantizer**) achieving $\sim$**19.7 ENOB**.
- Implemented **fixed and PRBS chopping techniques** to suppress flicker noise and recover ENOB degradation.

Power Converter Simulation Framework *NGSpice, GNU Octave, KiCad*

- Built a **modular NGSpice framework** for Buck, Boost, and Forward converters enabling reusable circuit simulation workflows.
- Developed custom **behavioral subcircuits** including PWM generators, non-ideal switch models, and gyrator-based transformer models.
- Modeled **switching losses, leakage inductance, and magnetic saturation** for realistic converter analysis.
- Validated transient and steady-state responses using **state-space modeling in GNU Octave**.

Technical Skills

Programming: C, C++, Python, MATLAB, Verilog HDL
Control & Power Electronics: FOC, IRFOC, Motor Drives, Power Electronics, State-Space Modeling, Model Predictive Control
Mixed-Signal Systems: $\Delta\Sigma$ ADC Design, Noise Modeling, Fixed/PRBS Chopping
Embedded Systems: Microcontrollers, DSP Systems, Real-Time Control, Digital System Design
Hardware Design: PCB Design, HV/LV Systems, Safety Electronics, Signal Integrity
Core Areas: Electric Vehicles, Electrical Machines, Control Systems, Embedded Systems

Achievements

- **6th Rank among 22 IITs** — Inter-IIT Tech Meet 14.0 (ISRO VLSI Challenge)
- **12th Rank in Engineering Design** and **24th Overall** — Formula Bharat 2025
- **AIR 7256** — JEE Advanced 2023
- **Top 5%** — Meta HackerCup 2024 (Practice + Round 1)

Leadership

- **Technical Secretary**, Hostel Executive Committee'23, Amber Hostel, IIT Dhanbad